

SpatialCliff: Mapping Where Scene Complexity Breaks Spatial Reasoning in an Open VLM

A Technical Report

Model under test: Qwen2.5-VL-3B-Instruct — an open 3B vision-language model, evaluated zero-shot.

Dataset: 480 procedurally-generated 2D scenes across 4 spatial-reasoning mechanisms × 6 complexity levels × 20 seeds, each with an exact ground-truth answer.

Question: Which spatial-reasoning capabilities survive scene complexity, and which collapse first?

1. Motivation

Multimodal models are asked to reason about space — "which object is closest to the blue diamond?", "what is partially hidden behind that square?" — in increasingly dense scenes. A 3B model handles these questions confidently on simple layouts. The question an application developer faces is:

Where does spatial reasoning quietly break as scenes get harder?

Simple scenes mask the failure: a model that fails on a cluttered 12-object scene may still score 1.0 on a 3-object scene. Aggregated benchmarks average over this variance. This project stress-tests four distinct spatial-reasoning mechanisms with a procedurally-generated scene corpus whose complexity is controlled along one axis at a time, and maps where each mechanism falls off a cliff.

2. Method

2.1 Scene corpus

Scenes are deterministic 2D top-down layouts rendered with PIL primitives. Four families probe distinct mechanisms:

family	task	what it requires	complexity axis
relpos	what color is the object to the left of the blue square?	locate a target, bind a property of a spatially-related object	object count 3→15
occlusion	what color is the object partially hidden behind a gray square?	reason through an occluder to recover a hidden property	occluders 1→5, objects 3→10
lookalike	which corner is the red circle with the blue dot closest to?	bind a small attribute to the right instance among identical distractors, then localize	identical circles 2→9

family	task	what it requires	complexity axis
nearest	what color is the object closest to the blue diamond?	relative-distance computation over the whole scene	object count 3→14

20 seeds × 6 difficulties × 4 families = **480 scenes**. Ground truth is computed from the layout itself, so a model failure is a reasoning failure, never an annotation artifact. The questions never reveal the answer, and each offers a closed set of options (color words, or corner labels) so scoring is unambiguous.

2.2 Protocol

Zero-shot, one scene at a time. Images are capped at 448×448 input pixels (the scenes are simple geometric layouts; the stress axis is scene complexity, not resolution). Output is normalized per family by an explicit, lenient matcher; a scene is correct iff normalized output equals ground truth.

2.3 Cliff detection

Accuracy per (family, difficulty) with Wilson 95% CIs. A **falloff (cliff)** is the first difficulty step (from simplest to hardest) where accuracy drops ≥15 points in one step **and** ≥20 points below the simplest scene.

3. Results

3.1 The headline: three mechanisms fall off cliffs, one is uniformly hard

Per-family accuracy over the complexity axis (20 seeds per point):

family	d0	d1	d2	d3	d4	d5	range	verdict
nearest	0.70	0.70	0.35	0.20	0.10	0.05	0.65	cliff @ d2
relpos	1.00	0.85	0.65	0.65	0.60	0.45	0.55	cliff @ d2
occlusion	1.00	0.90	0.95	0.75	0.75	0.55	0.45	cliff @ d3
lookalike	0.35	0.50	0.65	0.60	0.40	0.45	0.30	uniformly hard

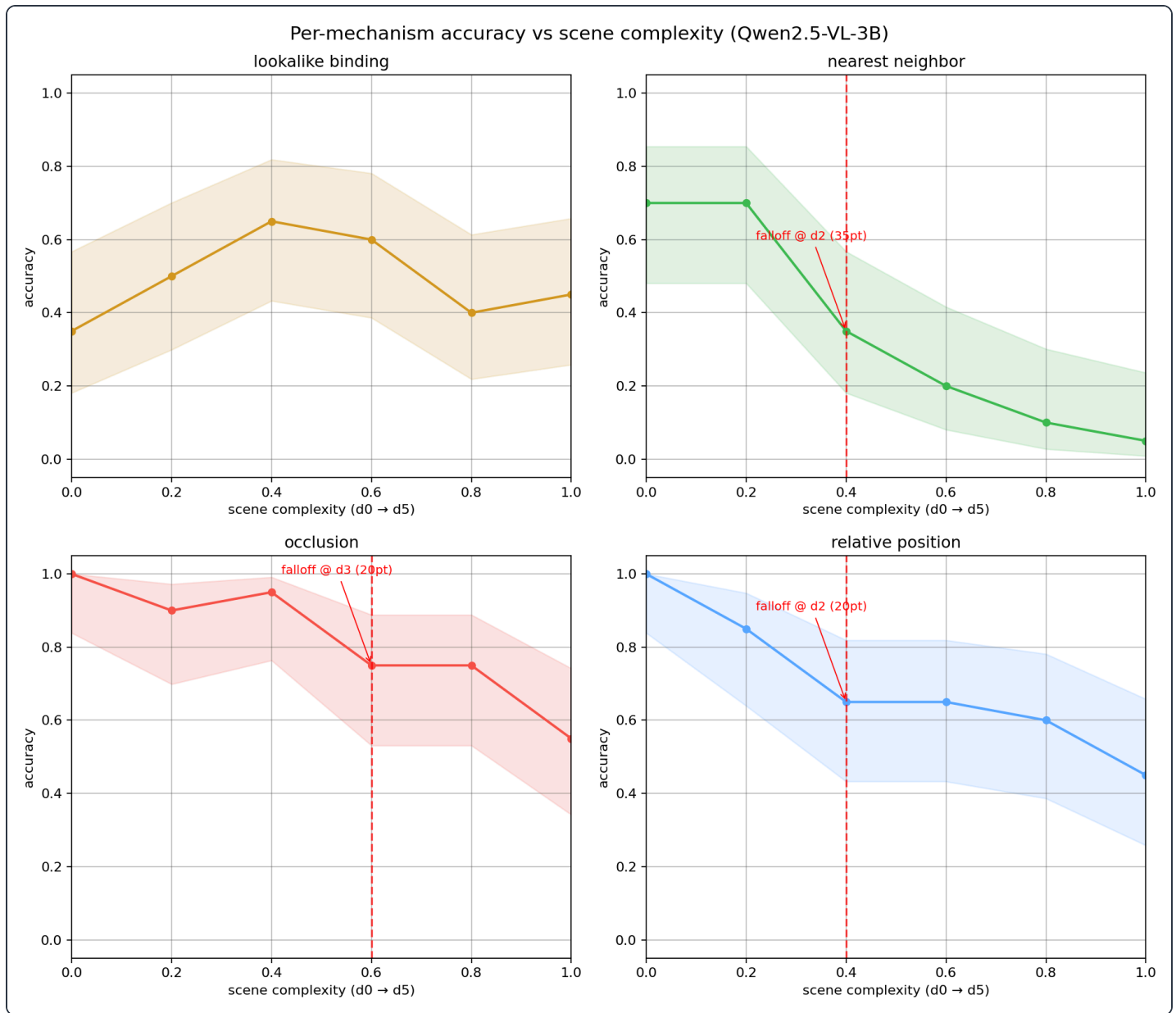


Figure 1

Figure 1 — Per-mechanism accuracy vs scene complexity. The red dashed line marks the detected falloff cliff.

3.2 Nearest-neighbor reasoning collapses first

nearest is the cleanest cliff: $0.70 \rightarrow 0.70 \rightarrow 0.35 \rightarrow 0.20 \rightarrow 0.10 \rightarrow 0.05$. The falloff fires at d2 (a 35-point drop in one step). Three objects are fine (0.70); four objects already drop to 0.35; fourteen objects leave the model at chance. Relative-distance computation degrades continuously once the candidate set passes ~3-4 objects — the model cannot keep more than a handful of pairwise distances in its working set.

3.3 Relative position degrades from a perfect baseline

relpos starts perfect (1.00 on 3 objects) and falls to 0.45 on 15 objects, with a cliff at d2. The mechanism is reliable when the target and its neighbor are prominent; as the scene fills with distractors, locating the correct neighbor becomes the binding bottleneck.

3.4 Occlusion reasoning survives mid-complexity, then falls

Occlusion holds at 0.90-1.00 through d2, falls at d3 (0.95 → 0.75), and reaches 0.55 at d5. Reasoning *through* an occluder is robust for a small number of objects; multiple occluders and clutter eventually overwhelm the hidden-object search.

3.5 Lookalike binding is uniformly hard — a mechanism boundary

Lookalike never exceeds 0.65 and stays near chance at the simplest scene (0.35). The model exhibits a **systematic bottom-quadrant attribution bias**: given several identical red circles, it disproportionately reports the target as bottom-left/bottom-right, even when the true position is top-right. This is not a complexity effect — it is a *mechanism boundary*: 3B-class models cannot reliably bind a small attribute to the correct instance among identical distractors, regardless of scene size. This bounds the claims we can make: attribute-instance binding is a hard floor for this model class, not a cliff.

4. Discussion

4.1 The spatial reasoning landscape

For an open 3B VLM, scene complexity trades away reasoning fidelity in a structured order:

1. **Nearest-neighbor is the most fragile** — the first mechanism to collapse, and it collapses completely (to 0.05).
2. **Relative position degrades from perfect** but never quite bottoms out (0.45 at maximum complexity).
3. **Occlusion reasoning is the most robust** — holds above 0.75 until the densest scenes.
4. **Attribute-instance binding is a hard floor**, not a cliff: uniformly near chance even at minimal complexity, with a systematic bottom-quadrant bias.

4.2 Implications for deployment

- **Scene-density budgets.** Applications that rely on nearest-neighbor judgments (e.g. "nearest obstacle", "closest target") should assume a hard limit of ~3 candidates for a 3B model; beyond that, reliability collapses.
- **Occlusion is safe mid-complexity.** Partially-hidden object reasoning holds to ~5 objects / 2-3 occluders, making it the most dependable spatial capability of the model class.
- **Don't trust aggregated accuracy.** All four mechanisms would show a "reasonable" average if pooled across difficulty; the cliffs only appear when accuracy is resolved per complexity level.

4.3 Failure modes, not just accuracy

The lookalike bottom-quadrant bias is the kind of finding aggregated benchmarks hide: the model does not fail randomly, it fails systematically, and the systematic error is itself a spec for where to invest (better attribute binding, better attention to small features).

5. Reproducibility

```
pip install -e .[gpu,paper,ui]
python scripts/build_scenes.py --seeds 20 --out data/scenes # 480 scenes
python scripts/run_sweep.py --scenes data/scenes --out data/sweep # GPU, ~4 min
python scripts/paper_facts.py --sweep data/sweep/sweep.json # derived claims
python scripts/render_figures.py --sweep data/sweep/sweep.json --figs docs/figures
python scripts/render_spatialcliff_paper.py # this report (HTML + PDF)
python -m spatialcliff ui --port 8000 # interactive console
```

`data/sweep/sweep.json` is the single source of truth: every number in this report traces to that table, and `scripts/paper_facts.py` re-derives the claims from it directly.

References

1. Wang et al. — Qwen2.5-VL Technical Report. *arXiv:2502.13923*.
2. Liu et al. — Visual Instruction Tuning. *NeurIPS 2023*.
3. Hudson & Manning — GQA: A New Dataset for Real-World Visual Reasoning and Compositional Question Answering. *CVPR 2019*.
4. Johnson et al. — CLEVR: A Diagnostic Dataset for Compositional Language and Elementary Visual Reasoning. *CVPR 2017*.
5. Li et al. — What If We Simply Repeat All Clues? A New Dataset for Probing Visual Recognition Capabilities of Multimodal Large Language Models. *arXiv:2410.06508*.
6. Yang et al. — SpatialVLM: Endowing Vision-Language Models with Spatial Reasoning Capabilities. *CVPR 2024*.